Dissipative Quantum Reference Frames: Non-Unitary Transformations Beyond the Operational Simultaneity Horizon

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Abstract

Recent advances in quantum foundations have generalized classical coordinate transformations to Quantum Reference Frames (QRFs), allowing reference clocks and spatial origins to exist in quantum superpositions. In standard formulations, switching between two QRFs A and B is modeled as an idealized, closed-system unitary transformation $\hat{U}_{A\to B}$. In this paper, we demonstrate that unitary QRF covariance holds strictly in the limit of naive classical point-locality, where spatial separation $\Delta\ell$ and environmental phase decoherence Γ_{tot} vanish. Extending the Coherence-Bounded Operational Locality framework, we show that any physical transformation between spatially separated quantum clocks requires a closed Einstein–Poincaré synchronization channel subject to round-trip signal delay $\Delta\tau_{\text{sig}} = 2\Delta\ell/(cN_{\text{eff}})$. Consequently, operational QRF shifts are governed by a dissipative completely positive trace-preserving (CPTP) map $\Lambda_{A\to B}$ weighted by the Spatial Locality Decay Factor $\mathcal{D}_{\text{loc}} = \exp(-\Delta\ell/\mathcal{L}_{\text{sim}})$. We derive a closed-form QRF Transformation Fidelity Bound $\mathcal{F}_{\text{QRF}}$ and prove that distributed quantum networks undergo a sharp architectural bifurcation at the Operational Simultaneity Horizon $\mathcal{L}_{\text{sim}}$. Finally, we analyze these results within a neutral four-dimensionalist ontology, showing how relational quantum mechanics remains covariant without requiring zero-latency signaling across spacetime.

1 Introduction: The Idealization of Unitary QRF Shifts

The formalism of Quantum Reference Frames (QRFs) has emerged as a cornerstone of modern quantum foundations and quantum gravity phenomenology [1, 2]. By internalizing the reference system—such as a quantum clock or spatial origin—within the total Hilbert space, QRF transformations reveal that physical properties like entanglement and superposition are frame-dependent, yet covariant under changes of quantum perspective.

In standard algebraic treatments, transforming the state ρ_A of a target system relative to clock A into the state ρ_B relative to clock B is represented by a unitary operator $\hat{U}_{A\to B}$:

$$\rho_B^{\text{ideal}} = \hat{U}_{A\to B}\,\rho_A\,\hat{U}_{A\to B}^{\dagger}. \tag{1}$$

While mathematically consistent within an abstract, timeless Hilbert space, Eq. (1) embeds a crucial physical idealization: it assumes that the phase and timing relationship between reference frames A and B can be established and maintained with zero signaling latency and zero thermodynamic noise. In earlier installments of this series (Papers 9 and 10), we identified this assumption as *naive classical point-locality*.

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When clocks A and B are separated by a proper physical baseline $\Delta\ell > 0$ within a noisy environment characterized by total phase decoherence rate $\Gamma_{\text{tot}} > 0$, operational frame alignment cannot occur instantaneously. Here, we bridge QRF theory with relativistic metrology by replacing $\hat{U}_{A\to B}$ with a coherence-bounded open-system channel.

2 Coherence-Bounded QRF Channel Formalism

Consider two d-dimensional quantum clock reference nodes A and B separated by a proper physical baseline $\Delta\ell$ in a waveguide or transmission medium of effective group refractive index N_{eff}.

2.1 Operational Simultaneity Horizon and Spatial Decay

As established in our prior work, establishing real-time phase synchronization between nodes A and B requires a two-way electromagnetic handshake with minimum round-trip propagation time:

$$\Delta\tau_{\text{sig}} = \frac{2\Delta\ell}{cN_{\text{eff}}}. \tag{2}$$

Given a target phase-locking threshold parameter $\rho_\tau \in (0, 1]$ and a total phase decoherence rate Γ_{tot} (in s^{-1}), active synchronization remains physically sustainable only within the **Operational Simultaneity Horizon** $\mathcal{L}_{\text{sim}}$:

$$\mathcal{L}_{\text{sim}} = \frac{cN_{\text{eff}}\rho_\tau}{2\Gamma_{\text{tot}}}. \tag{3}$$

The surviving fraction of operational phase-locking capacity at baseline $\Delta\ell$ is quantified by the dimensionless **Spatial Locality Decay Factor**:

$$\mathcal{D}_{\text{loc}} = \exp\left(-\frac{\Delta\ell}{\mathcal{L}_{\text{sim}}}\right) = \exp\left(-\frac{2\Gamma_{\text{tot}}\Delta\ell}{cN_{\text{eff}}\rho_\tau}\right). \tag{4}$$

2.2 The Dissipative QRF Channel ($\Lambda_{A\to B}$)

When an observer attempts an operational QRF transformation from A to B across $\Delta\ell$, the coherent unitary relational map $\hat{U}_{A\to B}$ succeeds only to the extent that the synchronization channel preserves phase coherence during $\Delta\tau_{\text{sig}}$. The uncompensated phase diffusion maps the remaining statistical weight onto the maximally mixed state $\mathbb{I}/d$.

Definition 1 (Coherence-Bounded QRF Channel)**.** *For a d-dimensional quantum state ρ_A defined relative to reference frame A, the physical operational transformation to a spatially separated reference frame B at baseline $\Delta\ell$ is given by the completely positive trace-preserving (CPTP) map $\Lambda_{A\to B}$:*

$$\Lambda_{A\to B}(\rho_A) = \mathcal{D}_{\text{loc}}\left(\hat{U}_{A\to B}\,\rho_A\,\hat{U}_{A\to B}^\dagger\right) + (1 - \mathcal{D}_{\text{loc}})\,\frac{\mathbb{I}}{d}, \tag{5}$$

where $\hat{U}_{A\to B}$ is the idealized unitary QRF operator and $\mathcal{D}_{\text{loc}} = \exp(-\Delta\ell/\mathcal{L}_{\text{sim}})$.

Theorem 1 (QRF Transformation Fidelity Bound)**.** *Let $\rho_A = |\psi_A\rangle\langle\psi_A|$ be a pure state in frame A, and let $|\psi_B^{\text{ideal}}\rangle = \hat{U}_{A\to B}|\psi_A\rangle$ denote its idealized unitary transform in frame B. The operational state fidelity $\mathcal{F}_{\text{QRF}} = \langle\psi_B^{\text{ideal}}|\Lambda_{A\to B}(\rho_A)|\psi_B^{\text{ideal}}\rangle$ is bounded by:*

$$\mathcal{F}_{\text{QRF}}(\Delta\ell, \Gamma_{\text{tot}}) = \frac{1}{d} + \left(1 - \frac{1}{d}\right)\exp\left(-\frac{2\Gamma_{\text{tot}}\Delta\ell}{cN_{\text{eff}}\rho_\tau}\right). \tag{6}$$

Proof. Substituting $\rho_A = |\psi_A\rangle\langle\psi_A|$ into Eq. (5) yields:

$$\Lambda_{A\to B}(\rho_A) = \mathcal{D}_{\text{loc}}|\psi_B^{\text{ideal}}\rangle\langle\psi_B^{\text{ideal}}| + (1-\mathcal{D}_{\text{loc}})\,\frac{\mathbb{I}}{d}.$$

Taking the expectation value with respect to $|\psi_B^{\text{ideal}}\rangle$ and using $\langle\psi_B^{\text{ideal}}|\mathbb{I}|\psi_B^{\text{ideal}}\rangle = 1$, we obtain:

$$\mathcal{F}_{\text{QRF}} = \mathcal{D}_{\text{loc}}(1) + (1-\mathcal{D}_{\text{loc}})\,\frac{1}{d} = \frac{1}{d} + \left(1-\frac{1}{d}\right)\mathcal{D}_{\text{loc}}.$$

Inserting Eq. (4) for $\mathcal{D}_{\text{loc}}$ completes the proof. □

Corollary 1 (Recovery of Unitary Covariance)**.** *Unitary QRF covariance ($\mathcal{F}_{\text{QRF}} \to 1$) is recovered asymptotically if and only if $\Delta\ell/\mathcal{L}_{\text{sim}} \to 0$, corresponding either to co-located frames ($\Delta\ell \to 0$) or noiseless clocks ($\Gamma_{\text{tot}} \to 0$).*

3 Architectural Bifurcation in Distributed Quantum Networks

Theorem 1 imposes a strict physical design constraint on distributed quantum computing and quantum internet architectures. Even in an idealized optical network with zero photon-loss attenuation, real-time quantum gate synchronization across separated nodes requires active QRF alignment.

Table 1: Architectural Regimes for Distributed Quantum Networks under $\Lambda_{A\to B}$

Network Tier	**Baseline Scale**	**QRF Fidelity** ($d=2$)	**Operational Protocol**
Monolithic Coherent Core	$\Delta\ell \ll \mathcal{L}_{\text{sim}}$	$\mathcal{F}_{\text{QRF}} \to 1.00$	Real-time unitary gate sync
Horizon Boundary Link	$\Delta\ell = \mathcal{L}_{\text{sim}}$	$\mathcal{F}_{\text{QRF}} \approx 0.684$	Threshold error-corrected sync
Wide-Area Quantum Link	$\Delta\ell \gg \mathcal{L}_{\text{sim}}$	$\mathcal{F}_{\text{QRF}} \to 0.50$	Post-selected Bell / teleportation

For qubit clocks ($d = 2$), once the fiber baseline exceeds $\mathcal{L}_{\text{sim}}$, $\mathcal{F}_{\text{QRF}}$ drops rapidly toward the classical random-guess floor of $1/2$. Consequently, large-scale quantum networks naturally bifurcate: active real-time phase-locked operations are restricted to local monolithic cores ($\Delta\ell \ll \mathcal{L}_{\text{sim}}$), while long-haul intercity links ($\Delta\ell \gg \mathcal{L}_{\text{sim}}$) must operate exclusively via post-selected entanglement correlations (S_{CHSH}), which remain immune to $\mathcal{D}_{\text{loc}}$ decay.

4 Philosophical Implications: Relational Covariance in Spacetime

Within a neutral four-dimensionalist (Block Universe) ontology, the worldlines of quantum clocks A and B and their local measurement events are fixed geometric structures on a Lorentzian manifold. Relational quantum mechanics and QRF theory rightly emphasize that quantum states describe correlations relative to a chosen physical reference system rather than absolute, observer-independent properties.

Our dissipative channel $\Lambda_{A\to B}$ clarifies the ontological status of QRF shifts in relativistic spacetime. Idealized unitary operators $\hat{U}_{A\to B}$ represent global mathematical symmetries across the four-dimensional manifold, analogous to passive coordinate diffeomorphisms. By contrast, an *operational* QRF transformation is a physical thermodynamic process occurring along timelike and null worldlines. Because electromagnetic synchronization is bounded by c and subjected to environmental entropy production Γ_{tot}, operational perspective-taking is inherently local and dissipative. Thus, relational covariance is preserved globally across spacetime without violating relativistic causality or requiring instantaneous physical frame-switching.

5 Conclusion

Unitary Quantum Reference Frame transformations rely on an implicit assumption of naive point-locality. By incorporating round-trip signaling delay $\Delta\tau_{\text{sig}}$ and phase decoherence Γ_{tot}, we have shown that physical QRF shifts across spatial baselines are governed by a dissipative CPTP channel $\Lambda_{A\to B}$ weighted by $\mathcal{D}_{\text{loc}}$. The resulting fidelity bound $\mathcal{F}_{\text{QRF}}$ defines the exact physical horizon separating coherent distributed quantum processors from post-selected entanglement networks.

AI Collaboration & Authorship Transparency Statement

This work is the result of human-AI scholarly collaboration. The core philosophical framework, conceptual architecture, operational definitions, and theoretical direction were conceived and authored by **Luís Pimenta Damásio** (ORCID: 0009-0001-7606-7557). Mathematical formalization, analytical verification, and LaTeX typesetting support were provided by **Gemini** (Google AI) under the direct supervision, review, and editorial authority of the human author.

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